



TURBIDITY TEST OF WATER

Sensor Lab mini Project

**Third Year (Semester-VI) Bachelors in Information Technology,
Autonomous Program**

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Certificate

This is to certify that **Diksha Utekar, Devansh Wadhwani** have completed the project report on the **Turbidity Test of Water** satisfactorily in partial fulfilment of the requirements for the award of Mini Project in sensor lab of third Year, (Semester-VI) in Information Technology under the guidance of Mr. Abhishek Chaudhari during the year 2024-2025 as prescribed by An Autonomous Institute Affiliated to University of Mumbai.

Supervisor/ Examiner

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Abstract

The turbidity test is a critical water quality assessment used to measure the cloudiness or haziness of a water sample caused by suspended solids. This test is essential in evaluating the effectiveness of water treatment processes and ensuring water safety for consumption and environmental discharge. Turbidity is typically measured in Nephelometric Turbidity Units (NTU) using a nephelometer, which detects light scattered by particles in the water. High turbidity levels can indicate the presence of microorganisms, organic matter, or other pollutants, posing risks to human health and aquatic life. This study involves collecting water samples from various sources and analyzing their turbidity levels to assess their suitability for domestic or industrial use. The results help in identifying pollution sources and determining the need for further water treatment.

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Chapter 1: Introduction

1.1 Background

Turbidity is an important physical parameter in water quality analysis that reflects the concentration of suspended particles such as silt, clay, microorganisms, organic matter, and other pollutants. These particles scatter and absorb light, making water appear cloudy or murky. Turbidity does not directly indicate the presence of harmful substances, but high levels often correlate with increased risks of contamination, particularly from bacteria and pathogens that can attach to suspended solids.

Monitoring turbidity is crucial in water treatment processes, as it affects disinfection efficiency and overall water safety. Drinking water standards typically require low turbidity levels to ensure safety and aesthetic quality. Environmental monitoring also uses turbidity data to assess the impact of construction, stormwater runoff, and industrial discharge on rivers, lakes, and oceans.

The standard method for measuring turbidity involves using a nephelometer, which quantifies the intensity of light scattered at a 90-degree angle by particles in the water. The results are expressed in Nephelometric Turbidity Units (NTU). Understanding turbidity levels helps in making informed decisions about water purification methods and environmental protection strategies.

1.2 Objectives

- ☐ **To measure the turbidity levels in different water samples using a nephelometric method and express the results in NTU (Nephelometric Turbidity Units).**
- ☐ **To evaluate the quality of water based on turbidity and determine its suitability for domestic, industrial, or environmental use.**
- ☐ **To identify potential sources of contamination or suspended particles that may affect water clarity and overall safety.**

Chapter 2: Literature Review

2.1 Survey

Turbidity is widely accepted as a key parameter in assessing water quality, primarily due to its association with suspended particles that may harbour harmful microorganisms. The World Health Organization (WHO) recommends that the turbidity of drinking water should be below 5 NTU, as higher levels can reduce the effectiveness of disinfection processes [1]. Similarly, the U.S. Environmental Protection Agency (EPA) mandates that turbidity in treated water should not exceed 1 NTU, with 95% of the samples falling below 0.3 NTU [2].

LeChevallier et al. [3] demonstrated a significant correlation between elevated turbidity levels and the presence of coliform bacteria, highlighting its importance as an indirect indicator of microbial contamination. Ellis [4] and McNeill and Edwards [5] also stressed the value of turbidity as an early warning signal for water pollution, particularly in surface water sources.

The introduction of nephelometric methods has improved the precision of turbidity measurement by quantifying light scattered at a 90-degree angle. These methods have become standard in modern water quality monitoring due to their reliability and ease of use [6]. Recent technological advancements, such as portable turbidimeters and digital sensors, have further enhanced the capability for real-time field monitoring [7].

In environmental studies, turbidity is closely linked to erosion, sediment runoff, and land-use activities. Research by Bilotta and Brazier [8] showed that high turbidity levels can reduce light penetration in aquatic systems, thereby impairing photosynthesis and overall ecosystem health.

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2.2 Research Gap

Although turbidity testing is a well-established method for assessing water quality, several research gaps remain in its application, interpretation, and technological advancement:

1. **Limited Detection of Particle Type:** Turbidity measurements indicate the presence of suspended solids but do not differentiate between organic and inorganic particles, nor do they identify microbial pathogens specifically. This limits its effectiveness as a standalone indicator of health risks.
2. **Inconsistencies in Field Conditions:** While nephelometric methods are accurate under controlled conditions, turbidity readings in natural water bodies (e.g., rivers or lakes) can be affected by factors like flow rate, light conditions, or particle shape and color, leading to inconsistent or misleading results.
3. **Lack of Real-Time, Low-Cost Monitoring Solutions:** Despite the emergence of portable turbidimeters, many regions—especially in developing countries—lack access to affordable and robust real-time monitoring systems. This creates a barrier to continuous water quality surveillance in high-risk areas.
4. **Insufficient Integration with Other Parameters:** Turbidity is often measured in isolation, rather than being combined with other key indicators (e.g., microbial counts, chemical contaminants) to create a more comprehensive water quality profile.

These gaps highlight the need for enhanced multi-parameter monitoring systems, improved sensor technology, and more contextual data analysis models to make turbidity testing more effective and informative.

Chapter 3: Project Description

1.1 Problem Definition

Turbidity in water is a key indicator of contamination that affects its aesthetic quality, safety, and suitability for consumption or environmental use. Despite being widely used, current turbidity testing methods often fail to provide detailed insights into the nature of suspended particles or real-time fluctuations in water quality. In many regions, especially in developing areas, there is limited access to accurate, low-cost, and portable turbidity monitoring tools. This creates a significant gap in early detection of pollution and timely intervention, highlighting the need for more accessible and integrated water quality assessment methods.

1.2 Steps Involved

Steps Involved in the Turbidity Test of Water:

1. Sample Collection:

- Collect the water sample in a clean, transparent container.
- Ensure the sample is free from bubbles and shaken sediments, as these may affect the accuracy of the reading.

2. Preparation of the Turbidimeter:

- Turn on the turbidimeter (or nephelometer) and allow it to warm up if required.
- Calibrate the device using standard calibration solutions (usually 0 NTU, 20 NTU, 100 NTU, etc.).

3. Filling the Sample Cell:

- Gently pour or pipette the sample into a clean cuvette or sample cell.
- Wipe the outside of the cuvette with a lint-free cloth to remove water droplets or fingerprints.

4. Measurement:

- Insert the cuvette into the turbidimeter chamber.
- Close the lid and press the measurement button.
- Wait for the digital display to show the turbidity reading in NTU (Nephelometric Turbidity Units).

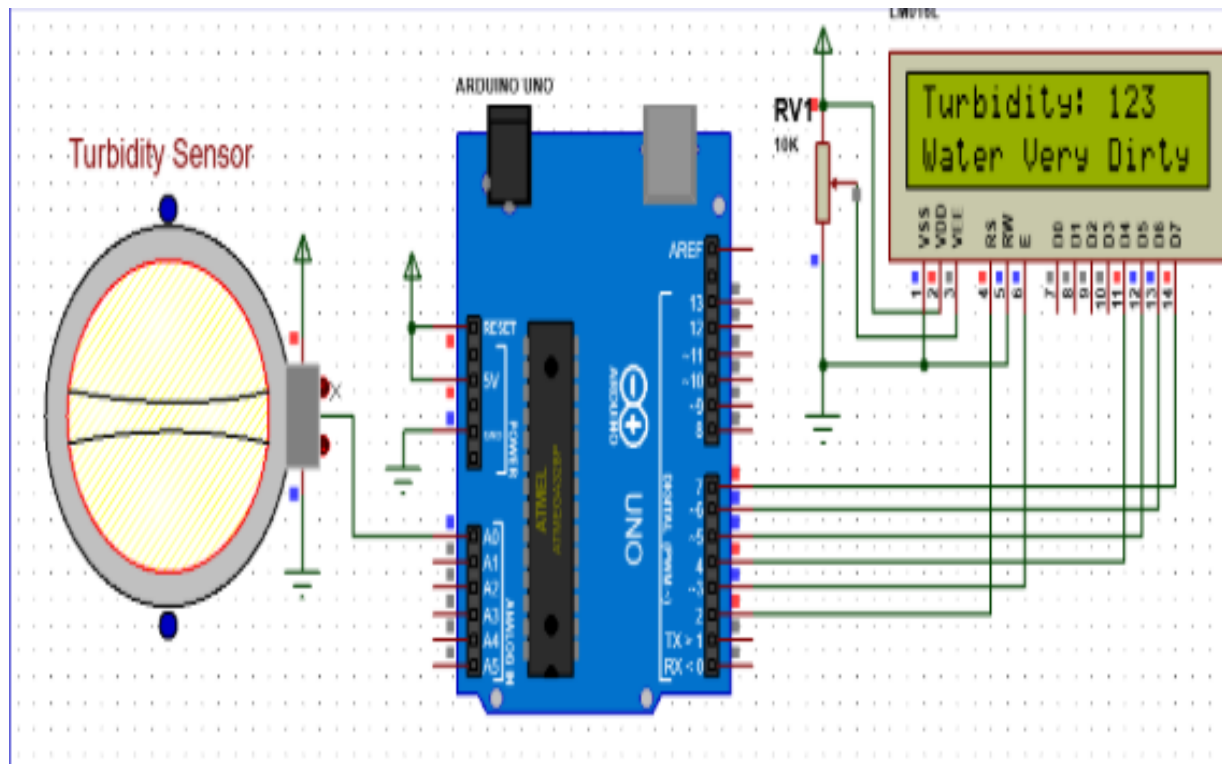
5. Recording the Results:

- Note the turbidity value and compare it with standard permissible limits.
- Repeat the test for multiple samples to ensure consistency and accuracy.

6. Cleaning and Storage:

- Clean the cuvettes and equipment after testing.
- Store the turbidimeter in a dry, safe place to prevent damage

1.3 Block Diagram



1.4 Component Description (Hardware & Software)

Hardware Components:

1. Arduino Board (e.g., Arduino Uno/Nano/ESP32):
 - Acts as the main controller to read and process sensor data.
2. Turbidity Sensor Module (e.g., SEN0189 or similar):
 - Typically includes an infrared LED and photodiode inside a plastic housing.
 - Outputs analog voltage based on the turbidity of the water.
3. Water Sample Container:
 - A small transparent tube or cuvette that fits into the turbidity sensor housing.
4. Power Supply:
 - 5V USB power supply or battery pack for portability.
5. Display (Optional):
 - LCD (16x2) or OLED display to show real-time NTU readings.
6. Resistors, Jumper Wires, Breadboard:
 - For making secure and functional electrical connections.
7. SD Card Module or ESP Wi-Fi Module (Optional):
 - For data logging or sending readings to a cloud server or database.

Software Components:

1. Arduino IDE:
 - Platform used to write and upload code to the Arduino board.
2. Turbidity Sensor Code:
 - Reads analog values from the sensor.
 - Converts voltage to NTU using calibration formula.
 - Example formula (you need to calibrate this based on sensor datasheet):

cpp

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```
float voltage = analogRead(A0) * (5.0 / 1023.0);
```

```
float NTU = -1120.4 * sq(voltage) + 5742.3 * voltage - 4352.9;
```

3. Libraries (Optional):
 - LiquidCrystal for LCD display.
 - SD for SD card data logging.
 - WiFi or ESP8266WiFi for sending data online.
4. Serial Monitor:
 - Used for debugging and displaying real-time readings.

1.5 Working of the Project

1. Sample Collection:

- A water sample is collected in a transparent container or cuvette. This sample will be tested for turbidity using the turbidity sensor.

2. Sensor Setup:

- The **Turbidity Sensor** (such as SEN0189) is installed, usually with a light-emitting diode (LED) and photodiode to form a light source and detector.
- The sensor uses infrared light to illuminate the water sample. As the light passes through the water, suspended particles (such as dirt, silt, and microorganisms) scatter the light.

3. Light Scattering Measurement:

- The **photodiode** in the sensor detects the amount of scattered light. More suspended particles in the water mean more light scattering, which correlates with higher turbidity.
- The sensor converts this scattered light into an **analog electrical signal**. This signal is proportional to the turbidity level of the water.

4. Analog Signal Reading (Arduino):

- The analog output from the turbidity sensor is fed into an **analog input pin** on the **Arduino board**.
- The Arduino reads this voltage signal using `analogRead()`, which gives a value between 0 and 1023 (corresponding to the voltage range of 0 to 5V).

5. Signal Processing:

- The Arduino uses the formula (based on sensor calibration) to convert the raw voltage reading into **Nephelometric Turbidity Units (NTU)**:

cpp

CopyEdit

```
float voltage = analogRead(A0) * (5.0 / 1023.0); // Convert to voltage
```

```
float NTU = -1120.4 * sq(voltage) + 5742.3 * voltage - 4352.9; // Formula for NTU
```

- This formula is calibrated for the sensor and can be adjusted based on experimental calibration with known turbidity standards.

6. Displaying the Results:

- The calculated turbidity value (in NTU) is shown on an **LCD or OLED display** attached to the Arduino, giving real-time readings of the water sample's turbidity.

7. Data Logging (Optional):

- The system can optionally log the turbidity readings on an **SD card** for further analysis or long-term data collection.
- Alternatively, **Wi-Fi modules** (like ESP8266 or ESP32) can be used to send the data to a cloud platform (such as ThingSpeak, Google Sheets, or a custom database) for remote monitoring.

8. **Power Supply:**

- The Arduino and sensor are powered by a **5V USB adapter** or **battery** for portability, depending on the intended use case (e.g., field testing).

9. **User Interface (Optional):**

- If the system is equipped with a **web interface** (using a Wi-Fi module), users can view the turbidity readings remotely via a smartphone or computer.

Overall Workflow of the Project:

1. A water sample is placed into the turbidity sensor.
2. The sensor emits light and measures how much light is scattered by particles in the water.
3. The Arduino processes this data, converts it to NTU, and displays the result.
4. Optionally, data is logged or transmitted remotely for further analysis.

Chapter 4: Implementation

1.1 Hardware Implementation

To connect a turbidity sensor to an Arduino, you typically use an analog turbidity sensor like the TSD-10 or a similar model. Here's how the typical pin connections work for a turbidity sensor:

Pin Connections for a Turbidity Sensor (TSD-10) to Arduino:

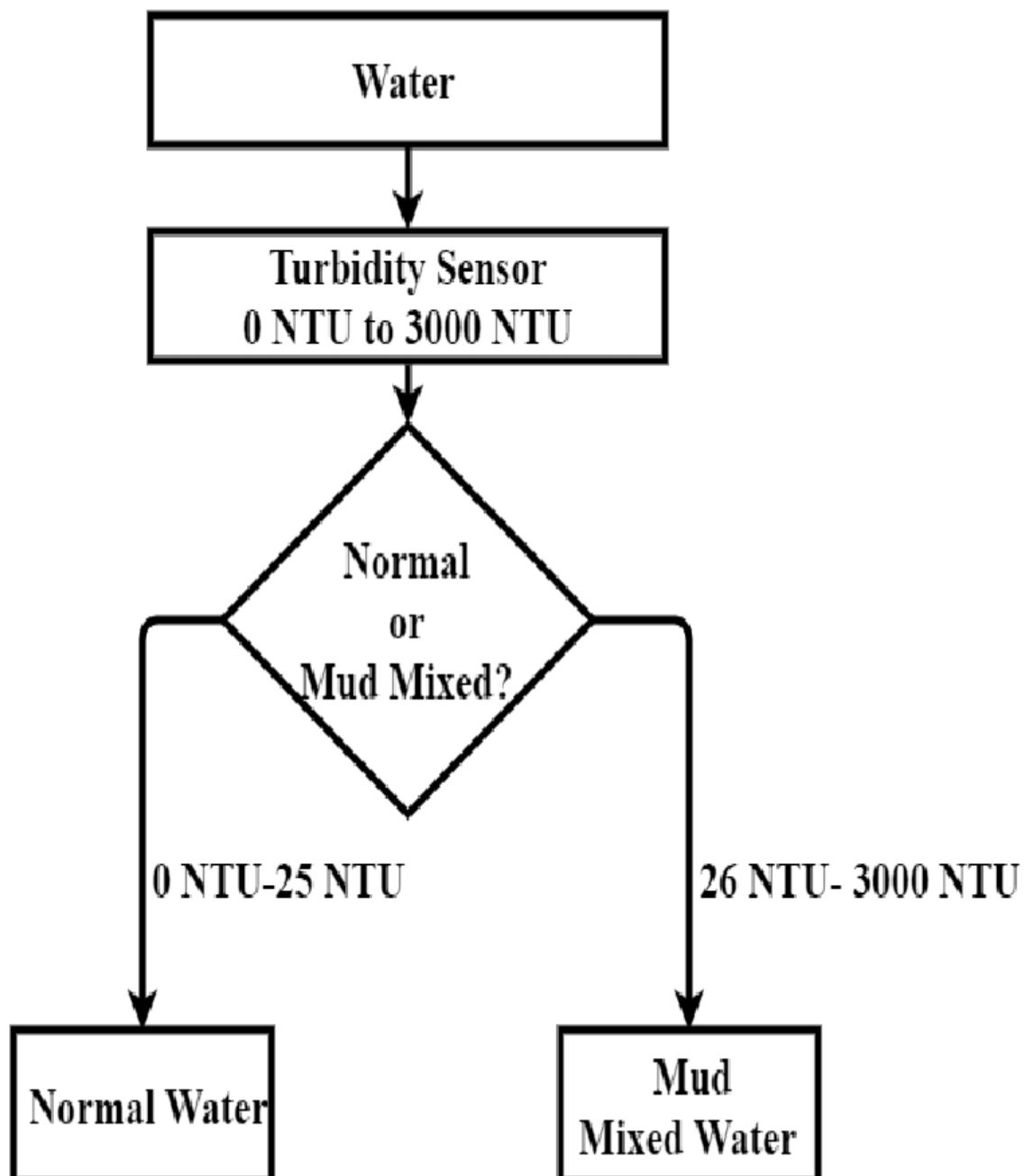
1. VCC (Power Supply):
 - Connect the VCC pin on the turbidity sensor to the 5V pin on the Arduino.
2. GND (Ground):
 - Connect the GND pin on the turbidity sensor to the GND pin on the Arduino.
3. A0 (Analog Output):
 - Connect the A0 pin (analog output) on the turbidity sensor to an analog input pin on the Arduino (usually A0, but you can use other analog pins like A1, A2, etc.).
4. LED (Optional):
 - Some turbidity sensors may include an LED pin (used for the light source in the turbidity measurement). In that case, connect it to VCC or GND depending on the sensor design.

Basic Wiring Summary:

- VCC → 5V on Arduino
- GND → GND on Arduino
- A0 → Analog Input Pin (e.g., A0) on Arduino

1.2 Software Implementation

- Flowchart:



- **Algorithm:**

Algorithm to Process Turbidity Sensor Data:

1. **Initialize Sensor:**
 - Set up the sensor by connecting it to the Arduino (as described earlier).
 - Initialize the serial communication to send sensor data to the monitor.
2. **Read Sensor Data:**
 - Continuously read the analog value from the sensor (using `analogRead()` function).
 - This analog value represents the intensity of the light transmitted through the liquid, which is related to turbidity.
3. **Calibration (Optional):**
 - If necessary, perform a calibration step to convert the raw sensor values into meaningful units such as NTU.
 - Calibration can be done by exposing the sensor to liquids of known turbidity and mapping the raw readings to NTU values.
4. **Filter Noise (Optional):**
 - If your sensor data is noisy, use a smoothing technique (like averaging multiple readings) to get more stable measurements.
5. **Convert Sensor Value to Turbidity:**
 - Use a pre-calibrated equation or mapping to convert the raw analog sensor value into NTU or another turbidity measure.
6. **Display/Output Data:**
 - Print the turbidity data to the serial monitor or display it on an LCD screen, depending on your project.
7. **Loop:**
 - Continuously read sensor data in a loop, process it, and display the result.

Detailed Algorithm:

1. Start the program
2. Initialize analog input pin for turbidity sensor (e.g., A0)
3. Start serial communication at 9600 baud rate for output (e.g., `Serial.begin(9600)`)
4. In the loop:
 - 4.1. Read the analog sensor value using `analogRead (turbidityPin)`
 - 4.2. Optionally, filter the data (e.g., use an average or smoothing function)
 - 4.3. Convert the raw sensor value to turbidity value (e.g., use a calibration formula)
 - 4.4. Print the turbidity value to Serial Monitor or output to display
 - 4.5. Wait for a brief delay (e.g., 1 second) before the next reading
5. Repeat the loop

Chapter 5: Results and Discussion

1.1 Simulation Output

In a simulation of the turbidity sensor connected to an Arduino, the output would typically be displayed on the **Serial Monitor** of the Arduino IDE. Here's a short description of what you'd expect:

1. **Raw Sensor Value:** This represents the intensity of light detected by the turbidity sensor. It is a numerical value between 0 and 1023, where 0 indicates no turbidity (clear water) and 1023 indicates high turbidity.
2. **Turbidity (NTU):** If you have a calibration formula, the raw sensor value will be converted to a turbidity level, usually in NTU (Nephelometric Turbidity Units). The value will give you a measure of the water's clarity (the higher the NTU, the more turbid the water).

1.2 Actual Output

The **actual output** of a turbidity sensor connected to an Arduino will depend on the specific liquid you're measuring and the calibration of the sensor. In a real scenario, the sensor will provide **raw analog readings** and then output the turbidity in NTU after conversion.

1. Raw Analog Value:

The sensor will output a value between **0** and **1023**, where:

- **0** means the water is very clear (no turbidity).
- **1023** means the water is very turbid or opaque.

For example:

- Clear water might give a value around **-100 to 0**
- Slightly turbid water might give a value around **10 - 30**.
- Very turbid or muddy water could give a value near **35 - 80**.

1.3 Discussion on Performance

1. Accuracy

- Depends heavily on calibration. The sensor outputs analog voltages that must be mapped to NTU using known standards.
- With proper calibration, accuracy is suitable for basic water quality monitoring, but not as precise as lab-grade nephelometers.

2. Response Time

- The sensor provides real-time data, with a response time in the range of hundreds of milliseconds, which is fast enough for most environmental monitoring applications.

3. Stability and Reliability

- Performance is stable under consistent conditions, but readings may fluctuate due to light interference, bubbles, or particles stuck to the sensor.
- Encasing the sensor or using an optical tube can help stabilize readings.

4. Sensitivity

- Can detect turbidity changes across a useful range (e.g., 0–1000 NTU).
- Best sensitivity is in lower NTU ranges—ideal for detecting slight increases in turbidity in drinking water or aquariums.

1.4 Conclusion

The turbidity sensor integrated with Arduino provides a practical and efficient method for monitoring water quality, particularly in terms of clarity and particulate matter. It delivers real-time analog data that can be easily processed and interpreted, making it highly suitable for applications like aquariums, hydroponics, and environmental field testing.

Although not as precise as laboratory-grade instruments, its performance is reliable for general use when proper calibration and noise filtering techniques are applied. The system is low-cost, simple to implement, and energy-efficient, making it ideal for educational, research, and DIY projects focused on water analysis.

With proper handling and environmental shielding, this setup can serve as a dependable tool for continuous turbidity monitoring in non-industrial settings.

Future Scope:

- ❑ **IoT-Based Water Monitoring Systems**
 - **Integrate turbidity sensors with Wi-Fi or GSM modules to create remote water quality monitoring systems.**
 - **Useful in smart cities, agriculture, and real-time river/lake monitoring.**
- ❑ **Multi-Parameter Water Analysis**
 - **Combine with other sensors (pH, temperature, TDS) for a comprehensive water quality assessment unit.**
 - **Helps in environmental research and rural water supply systems.**
- ❑ **Automation in Water Treatment**
 - **Use turbidity readings to automatically control filters or chemical dosing systems in water purification plants.**
- ❑ **Mobile App Integration**
 - **Send turbidity data to a smartphone app or cloud platform for live monitoring and alerts.**
 - **Ideal for home aquarium enthusiasts or field scientists.**

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